# Effects of Medicaid Expansion on Insurance Status, Emergency Department Payers, and Rural Hospital Finances*

John F. Martin
Harvard Kennedy School

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**Abstract**

The 2010 Patient Protection and Affordable Care Act (ACA) allowed states to expand Medicaid eligibility to nearly all low-income adults beginning in 2014. Its coverage gains are well-documented, but less is known about who pays for uninsured care and whether expansion removes this burden. I leverage the staggered timing of state adoption in a difference-in-differences design and follow expansion from insurance coverage to emergency department (ED) payers, and finally to the finances of rural hospitals. First, I find that expansion reduced the uninsured rate by roughly 26–29%, raised Medicaid enrollment by 32–42%, and reduced private insurance enrollment by about 3%. Second, total ED visits did not change detectably, but Medicaid-paid visits rose by more than half, while self-pay and no-charge visits fell by over 40%. Finally, I follow this payer shift to roughly 2,300 rural hospitals from 2011–2019. Expansion cut charity care for uninsured patients by more than half, and their bad debt by more than a quarter. This occurred with little change in inpatient volume. Part of this relief was offset by a rise in the unreimbursed cost of treating Medicaid patients, leaving net relief of about half a million dollars a year for the average hospital. In short, at the emergency department and rural hospitals, expansion changed how care is financed far more than the volume of care provided.

## 1 Introduction

The Affordable Care Act, passed in 2010, allowed states to expand Medicaid, the joint federal–state insurance program for low-income people, beginning in 2014. Medicaid is a costly and impactful program. In 2023, 32% of U.S. residents were enrolled in Medicaid or the Children's Health Insurance Program (CHIP) at some point during the year; Medicaid accounts for roughly 10% of federal outlays, and Medicaid and CHIP together account for about 18% of national health expenditures (Medicaid and CHIP Payment and Access Commission (MACPAC), 2024). Expanding Medicaid, which is an opt-in decision by states, offers coverage to nearly all adults with incomes below 138% of the federal poverty level (FPL). Prior to expansion, states were only required to cover groups such as pregnant women, children, and disabled individuals, each subject to income guidelines. By the end of 2014, 26 states and DC had expanded Medicaid. By 2019, this number had increased to

*jfmartin@hks.harvard.edu. Thank you to Brittany Street, Associate Professor of Economics and Public Policy at the University of Missouri, for supervising this project and to the McNair Scholars Program for funding this research.

33 states. While expansion raising coverage is well-established, less is known about whether new spending paid for additional care, or instead for already-provided indigent care. This paper follows expansion from insurance coverage to the emergency department, and then to the finances of rural hospitals, to answer that question.

Medicaid expansion's effects can be measured by how it impacts the uninsured rate in a state. If effective, I expect several consequences. First, there would be a decreased financial barrier for low-income populations to obtain coverage. Subsequently, this results in greater access to healthcare for a population that previously had little to no access. Simply increasing access, though, does not necessarily result in increased enrollment. For example, even though many students are eligible to receive significant financial aid to attend college, many students and parents do not claim that aid because the process of filling out the FAFSA form is challenging (Kofoed, 2017). On the other hand, the expected benefit of coverage for this population, who otherwise would not have health insurance, may outweigh the costs of enrolling, as some of their diseases and afflictions are preventable (e.g., asthma, diabetes). The first purpose of this paper is to study empirically whether individuals respond to Medicaid expansion by enrolling, thus decreasing the uninsured rate.

Proponents of Medicaid expansion argue that it allows low-income individuals better access to preventive care. For a state, the costs of expanding are covered almost entirely by the federal government, resulting in minimal additional costs, a neutral budget impact, or even savings. Medicaid expansion could also shift patients from indigent care to Medicaid, resulting in savings for hospitals, in particular those that cannot turn patients away. Opponents, on the other hand, caution that while costs for states are not yet high, states bear a growing share. They worry that an increase in healthcare access may result in overcrowded hospitals, longer waits, and a lower quality of care. These opponents often prefer a more market-based approach to health care. There is an additional debate on how expansion might affect hospitals. One side expects expansion to relieve hospitals of the cost of treating patients who cannot pay, whereas the other expects it to strain hospitals with new demand. The second purpose of this paper is to test which of these occurred.

I use state-level data from 2001–2019 to study how Medicaid expansion affects two bodies of outcomes, namely insurance coverage and the cost of care. I first consider the uninsured rate (defined as the share of a state's population without health insurance, the private health insurance (PHI) enrollment rate (the share of the population enrolled in private coverage), and the Medicaid enrollment rate. In addition, I consider the Medicare enrollment rate as a placebo. Even though Medicare enrollment is related to health insurance, I do not expect it to respond, as Medicare eligibility is unrelated to the act of expanding. Then, I consider emergency department (ED) visits per resident, broken down into total, Medicaid, Medicare, self-pay/no-charge, and privately-insured visits. Finally, I consider how these effects materialize at the hospital level. Using a panel of roughly 2,300 rural hospitals, I study charity care, bad debt, Medicaid shortfalls, operating margins, inpatient volume, net patient revenue, and operating expense.

To distinguish the effect of Medicaid expansion from confounding factors, I exploit within-state variation in adopting and non-adopting states through a staggered difference-in-differences research

design, focusing primarily on within-region comparisons through region-by-year FE. I compare changes in outcomes within a state after it expands to contemporaneous changes in states that have not expanded. My crucial assumption is that, in the absence of Medicaid expansion, adopting states would have experienced changes in outcomes similar to non-adopting states within their region.

The data enable me to test this assumption in a number of ways. First, graphical evidence and event-study estimates show that the uninsured rate and ED visits by payer in adopting and non-adopting states did not diverge in the years prior to adoption. Additionally, my findings are robust to the inclusion of the poverty rate, the unemployment rate, and median household income; and to contemporaneous Medicare enrollment, which acts as a measure of general health insurance trends; and to state-specific linear time trends. Models are also robust to estimators designed for staggered adoption (Callaway and Sant'Anna, 2021; Wing et al., 2024) as well as randomization inference. At the hospital level, I use the same design, comparing rural hospitals in adopting and non-adopting states before and after expansion.

My results indicate that expanding coverage does not change Medicare enrollment. In contrast, I find that expanding states experience a decline of roughly 29% in the uninsured rate and an increase of up to 42% in Medicaid enrollment. I estimate that expanding states experience a decrease in the PHI enrollment rate of about 3%. These findings indicate that access matters for low-income health insurance enrollment. Most of the movement into Medicaid comes from low-income individuals who would otherwise have been uninsured. Expanding does result in somewhat lower PHI enrollment, which could be due to several reasons. For example, in states that did not expand, adults between 100 and 138% of the FPL qualified for subsidized private plans on the ACA Marketplaces, while in expanding states the same adults qualified for Medicaid. Some low-income workers may also find Medicaid cheaper than their employer's plan. However, the movement out of PHI, about 1.7 percentage points of the population, is about half as large as the movement out of uninsurance.

When discussing government involvement in healthcare, it is relevant to weigh the expected benefit of more insured low-income individuals against the corresponding expected costs. I find negligible effects on total and Medicare ED visits, and a modest decline in privately insured visits. There is, however, a 58% increase in Medicaid-paid ED visits and a 44% decrease in self-pay/no-charge ED visits. This implies a shift from self-pay or indigent care to Medicaid at the emergency department, with no detectable change in the amount of emergency care delivered.

At rural hospitals, expansion reduced the cost of charity care for uninsured patients by about 59%. Non-Medicare bad debt fell by about 29%. Inpatient volume and occupancy changed little. Relief was partly given back, as the unreimbursed cost of treating Medicaid patients rose by about 20%. This may occur because Medicaid pays less than the cost of care. Net relief was about $ 545,000 a year for the average rural hospital in an expanding state. Operating margins rose by about 2.2 percentage points, but this estimate is sensitive to pre-existing trends, so I treat it only as suggestive. The results describe a policy that, at the emergency department and rural hospitals, changed who pays for care more than how much care is delivered.

## 2 Literature Review

These findings contribute to three bodies of literature. The first focuses on barriers to enrollment in public programs among eligible people. As discussed above, Kofoed (2017) finds that students forgo an estimated $9,741.05 in aid on average by not completing the FAFSA form, a fact often attributed to the complexity of the form. Health insurance is no exception. Shepard and Wagner (2025) find that when Massachusetts stopped automatically enrolling its poorest residents in free coverage, new enrollment fell by a third, with the largest losses among younger, healthier, and poorer people. Finkelstein et al. (2019) find that low-income adults in Massachusetts are willing to pay less than half of what it costs to cover them, which they attribute in part to the uncompensated care the uninsured already receive. Enrollment in the Supplemental Nutrition Assistance Program (SNAP), a government assistance program focused on improving nutrition in lower-income households, is closely tied to economic conditions. Ganong and Liebman (2018) find that rising unemployment explains most growth in SNAP enrollment after the Great Recession. Whether eligible people utilize benefits is therefore an empirical question across a wide menu of policies.

The second strand concerns the direct effects of Medicaid expansion. There is a broad literature examining Medicaid expansion and its effects on a range of outcomes, such as health status and economic consequences (Antonisse et al., 2018). The most direct evidence concerns coverage. Courtemanche et al. (2017) find that in its first year the ACA raised insurance coverage by about twice as much in expansion states as in non-expansion states. However, Miller and Wherry (2017) find that by the second year, the uninsured rate among low-income adults had fallen by 8.2 percentage points more in expanding states. Kaestner et al. (2017) also find large gains in Medicaid coverage among low-income adults with little change, with little change in employment status. Some studies focus on health outcomes as indicators of policy effectiveness. Cole et al. (2017) find that expansion not only increased Medicaid coverage and decreased the uninsured rate (which is often used to quantify access to healthcare), but also improved quality of care. Saloner and Maclean (2020) analyze substance use disorder (SUD) treatment, finding that admissions rose steadily after expansion, and the share of admissions paid by Medicaid rose by 23%. Related effects have been studied for several other health outcomes (Polsky et al., 2017; Simon et al., 2017; Wen et al., 2015, 2017).

A common fear surrounding access to care for low-income individuals is ex-ante moral hazard, which is the worry that newly-insured individuals will take less care of their health once they are insured. Simon et al. (2017) study low-income childless adults, finding that while expanded coverage reaches its targeted population, there is no evidence that ex-ante moral hazard increased. Another fear is that, as demand for healthcare exceeds supply, expanding Medicaid will increase wait times. Allen et al. (2022) find that ED wait times in expanding states increased by an estimated 3.1 minutes. Evidence from earlier expansions is more reassuring for existing enrollees, as Ndumele et al. (2014) find no evidence that state expansions between 2000 and 2009 eroded access to care, and also did not increase ED use among enrolled adults already. Whether coverage raises ED use at all is also debated. Taubman et al. (2014) find that Medicaid raised ED visits by 40% among winners of Oregon's coverage lottery, while Nikpay et al. (2017) find that the 2014 expansions shifted ED visits

from the uninsured toward Medicaid, with a small rise in total visits. Freedman et al. (2017) find the same shift in inpatient discharges with no change in total hospitalizations.

There is also extensive research on the economic implications of Medicaid expansion. A feared consequence is that public coverage crowds out private insurance. Frean et al. (2017) find little evidence that the increase in Medicaid coverage under the ACA came at the expense of private coverage, while Semprini (2023) estimates that expansion reduced private coverage among adults by 1.5 percentage points. In this paper, I provide my own estimate of expansion's effect on private health insurance enrollment, a decline of about 1.7 percentage points, which is more in line with Semprini's estimate.

The third strand concerns hospitals. Hospitals cannot turn away emergency patients, so they absorb much of the cost of treating the uninsured. Garthwaite et al. (2018) estimate that each additional uninsured person costs hospitals roughly $800 a year in uncompensated care. Finkelstein et al. (2019) estimate that about 60% of Medicaid spending in Oregon was a transfer to providers of uncompensated care, so only the remaining 40% paid for additional care. Dranove et al. (2016) find that uncompensated care fell for hospitals in expanding states, but not hospitals in non-expanding states. Blavin (2016) finds the same in cost report data, along with operating margins about 1.1 percentage points higher; conversely, Nikpay et al. (2016) find that uninsured hospital stays fell sharply in expanding states in 2014. Duggan et al. (2022) find that expansion in California raised hospital revenue, driven by higher reimbursement per patient. Rural hospitals have attracted particular attention. Kaufman et al. (2016) document that rural closures rose sharply after 2010, and closing hospitals were more likely to be in the South and in states that had not expanded Medicaid. Lindrooth et al. (2018) find that expansion is associated with better hospital financial performance and a lower likelihood of closure, especially in rural markets.

I study financial and operational rural hospital outcomes. I extend the payer shift at the emergency department documented by Nikpay et al. (2017) through 2019 and across every adoption cohort. I add to this literature by connecting that shift to rural hospital finances, separating the relief into charity care, bad debt, and Medicaid shortfalls.

# 3 Data

## 3.1 State Panel

Outcome data for the uninsured rate come from the U.S. Census Bureau, from the Current Population Survey through 2012 and the American Community Survey from 2013 onward, and cover all 50 states and the District of Columbia from 2001–2019; the rate measures uninsured individuals of all ages as a percentage of a state's civilian noninstitutionalized population. Outcome data on Medicaid, Medicare, and private health insurance enrollment are from the Centers for Medicare & Medicaid Services (CMS) National Health Expenditure Accounts, which count enrollees of all ages by state of residence. Data on when each state expanded Medicaid come from the Kaiser Family Foundation (KFF), which records the date of expansion for each state; Figure 1 shows when each

state expanded. State population is obtained from the U.S. Census Bureau. ED visits by expected payer come from the Healthcare Cost & Utilization Project (HCUP) for the 36 reporting states, in an unbalanced panel of adult visits from 2006 through 2019.[1] Inpatient stays for asthma by expected payer come from HCUP for 2003–2017.

I use these data to examine whether expanding access to healthcare has an effect on insurance status. An individual is generally eligible for Medicare at age 65 or older.[2] As discussed previously, Medicaid eligibility depends on state rules, although it typically depends on factors such as income, age, and disability status. Anybody may buy private health insurance, although many receive it through an employer. All three programs require that individuals enroll. That is, an individual does not automatically have Medicaid, Medicare, or PHI because they fit the criteria for eligibility.

Given an individual is uninsured and eligible, one might expect a rational individual to enroll in Medicaid, as the expected benefit of being insured outweighs the cost of participating. The measure of uninsurance I use is the percent of individuals in a state who are not insured. Finally, I obtained the poverty rate from the U.S. Census Bureau as a time-varying covariate, along with median household income from the Census Bureau's Small Area Income and Poverty Estimates (SAIPE) and the unemployment rate from the Bureau of Labor Statistics' Local Area Unemployment Statistics. Table 1 reports summary statistics for these variables.

### 3.2 Rural Hospital Panel

The second half of the paper uses a hospital-level panel built from three sources. The first is the Healthcare Cost Report Information System (HCRIS), the annual Medicare cost report that every certified hospital is required to file. Worksheet S-10 of the report gives the cost of uncompensated care, the cost of charity care for uninsured patients, the cost of non-Medicare bad debt, and the unreimbursed cost of treating Medicaid patients, each of which I measure in logs of 2019 dollars. Worksheet S-3 contains beds, inpatient days, and discharges. Worksheets G-2 and G-3 give patient revenue and operating expense, from which I construct operating margins. The second source is the CMS Provider of Services (PoS) file, an annual census of certified facilities, which records whether each hospital is rural or a critical access hospital, and its ownership status. The third is the Census Bureau's Small Area Health Insurance Estimates (SAHIE), which contains the share of uninsured adults below 138% of the Federal poverty line (FPL). County poverty and median household income come from SAIPE, and county unemployment comes from the Local Area Unemployment Statistics (LAUS). All dollar figures are in 2019 dollars.

The sample is restricted to rural short-term acute or critical-access hospitals that filed a cost report in 2011–2013, observed through 2019, which is roughly 2,300 hospitals and 20,100 hospital-years across 48 states.[3][4]

[1] I exclude Illinois due to gaps in its HCUP reporting.

[2] Individuals with a qualifying disability, End-Stage Renal Disease, or ALS are also eligible for Medicare.

[3] Form 2552-10 was phased in for fiscal years beginning in May 2010, thus the panel begins in 2011 at the earliest. Rural status is recorded annually, so I classify each hospital by its status in 2011–2013, which keeps any hospital from entering or leaving the sample in response to expansion.

[4] A hospital that converts to critical access status, or that is involved in a merger, is issued a new Medicare

# 4 Methodology

## 4.1 Staggered Difference-in-Differences

To distinguish the effect of Medicaid expansion from confounding factors, I leverage the rollout across all states, exploiting within-state variation from 2014–2019. Because states adopted at different times (and some never adopted), I use a staggered difference-in-differences research design to compare changes in adopting states with changes in non-adopting states, focusing primarily on within-region comparisons. I estimate ordinary least squares (OLS) and weighted least squares (WLS) models of the form

$$Outcome_{st} = \beta_0 \cdot (Treat \times Post)_{st} + \beta_1 \cdot X_{st} + c_s + u_t + W_{rt} + \epsilon_{st} \tag{1}$$

where $Outcome_{st}$ is the outcome for state $s$ in year $t$, such as the uninsured rate, the Medicaid, Medicare, or PHI enrollment rate, or ED visits per resident by payer. $(Treat \times Post)_{st}$ equals the proportion of year $t$ in which state $s$'s expansion was in effect. $X_{st}$ is a vector of time-varying controls, which in my specifications are the state poverty rate, unemployment rate, and log median household income. Because outcomes are in logs, I report each coefficient $\beta_0$ in the text as a percent change $100 \times (e^{\beta_0} - 1)\%$.

I also examine whether time-varying determinants of uninsured rates are orthogonal to within-state variation in Medicaid expansion. $c_s$ are state FE, which account for observable and unobservable time-invariant differences across states. For example, there may be differences in baseline poverty rates or underlying health conditions. $u_t$ are year FE, which account for observable and unobservable shocks from year to year that affect all states the same. For example, each state was affected by a presidential election. To the extent each state was affected the same by these events, they are accounted for by $u_t$. In some specifications, $W_{rt}$ is included and accounts for region-by-year FE, which allow year-to-year shocks to be region-specific. For example, the Great Recession may affect all states in the Southeast the same but is allowed to affect states differently in New England versus the South. This is relevant because adoption was regionally concentrated. Additionally, in some specifications, I weight by population, as treating states equally gives a resident of a small state the same influence as a large state. I cluster standard errors by state.

The identifying assumption is that, absent Medicaid expansion, adopting states would have experienced changes in outcomes similar to non-adopting states. I can relax this assumption in specifications that include region-by-year FE, where the assumption becomes that in the absence of Medicaid expansion, an adopting state would have experienced changes in outcomes similar to a non-adopting state within its region. My data allow me to examine and relax these assumptions in several ways. First, I look graphically for evidence that the two groups diverged before treatment. Second, I estimate event-study versions of Equation 1 that replace the single treatment variable with indicators for each year relative to adoption. This allows me to observe whether states diverge

certification number. I thus link predecessor and successor numbers to construct a facility-level panel.

before opting in, which would violate the identifying assumption. Third, I estimate a specification that adds an indicator for two years prior to adoption.

The identifying assumption implies that factors related to outcomes should not systematically change once a state has adopted, since that would imply that the state would have diverged regardless of adoption status. In turn, I use Medicare enrollment as a measure of general health insurance trends to test this. It is possible that Medicaid expansion could affect Medicare enrollment, but any such effect would most likely be second-order at best, so I can include Medicare enrollment as a control to pick up differential trends in health insurance. This same reasoning allows a placebo/falsification test. Since expanding affects only the criteria for Medicaid eligibility, I expect no effect on Medicare enrollment. Finally, I allow each state to follow its own trend via state-specific linear time trends.

Because adoption is staggered, a two-way FE regression can use already-treated states as controls for later-treated ones, which biases the estimate when effects grow over time (Goodman-Bacon, 2021). I therefore also estimate a stacked difference-in-differences model (Wing et al., 2024) and Callaway and Sant'Anna (2021)'s estimator. I summarize these results in section 6.

### 4.2 Hospital-Level Design

At the hospital level, I estimate the same difference-in-differences design, replacing state FE with hospital FE. For hospital $h$ in state $s$ and year $t$, I estimate

$$Outcome_{hst} = \beta_0 \cdot Post_{hst} + \beta_1 \cdot X_{ct} + c_h + u_t + \epsilon_{hst} \tag{2}$$

where $Post_{hst}$ is the share of hospital $h$'s cost-reporting period that falls on or after state $s$'s expansion date, $X_{ct}$ contains the poverty rate, unemployment rate, and log median household income of the hospital's county $c$, $c_h$ are hospital FE, and $u_t$ are year FE. In a second version, I replace year FE with region-by-year FE. I cluster by state.

The identifying assumption for Equation 2 is that, absent expansion, rural hospitals in adopting states would have trended similarly to rural hospitals in non-adopting states. I again examine these trends by observing the raw data, and then with event-study versions of Equation 2. I estimate this specification on two groups of hospital outcomes. The first is payer mix, which covers the uninsured rate among low-income adults in the hospital's county, the cost of charity care for uninsured patients, non-Medicare bad debt, and the unreimbursed cost of Medicaid patients. The second group is operating conditions, namely operating margin, log discharges, occupancy, log net patient revenue, and log operating expense.

## 5 Results

### 5.1 Insurance Status

Figure 2 shows log uninsured rates over time for adopting and non-adopting states, by year of adoption. For example, panel (b) shows log uninsured rates for states that expanded Medicaid in

2015 (Alaska, Indiana, and Pennsylvania) and compares them to states that had not adopted by 2019. Two clear patterns emerge from these figures. First, the uninsured rates of adopting states follow trajectories similar to those of non-adopting states prior to adoption. This suggests that had expanding states not adopted, they would have trended similarly to those that did not (parallel trends assumption). Second, in the years following adoption, uninsured rates fall in adopting states relative to the control group, for every adoption cohort. The difference is clearer the longer a state has been adopted. While the data only span 2001–2019, expanding states clearly experience a relative decrease in uninsured rates.

A formal analysis of how Medicaid expansion affects uninsured rates is shown in Table 2. Panel A reports unweighted OLS on the log outcome, Panel B weights by the state's average population over the sample period, and Panel C reports a negative binomial model of the implied number of uninsured residents, with state population as the exposure. The first column is a standard two-way FE benchmark with state FE and year FE. The second column adds region-by-year FE, which allows shocks in a given year to be regional. This is my preferred model. The remaining specifications test whether my preferred model is a reasonable choice. If it is, I expect these coefficients to be relatively stable around my preferred model's estimate. The third column adds the poverty rate, the unemployment rate, and log median household income as time-varying controls to test whether the model is robust to economic conditions, and these controls remain in every later column. The fourth column adds an indicator for the two years before adoption. The fifth column re-estimates the fourth without the ten states that covered low-income adults before 2014, either through early ACA expansions (California, Connecticut, the District of Columbia, Minnesota, New Jersey, and Washington) or through earlier state programs (Delaware, Massachusetts, New York, and Vermont), since part of their treatment falls before their 2014 expansion date. The sixth column includes contemporaneous Medicare enrollment as a control. This specification tests whether changes in the uninsured rate are driven by general health insurance trends rather than by Medicaid expansion. The seventh and final specification adds state-specific linear time trends, which allow states to trend differently over time.

My preferred model indicates a highly significant decline of roughly 29% in the uninsured rate as a result of expanding in the unweighted model, and a 26% decline when weighting by population. These estimates stay relatively stable, between 27 and 30% in columns C through F across both weighting schemes, which is what I expect if the identifying assumption holds. The indicator for the two years before adoption in column D is small, at $-4.1\%$ unweighted (significant at the 10% level) and $-6.1\%$ weighted (significant at the 5% level). Column E shows that part of this comes from the states with early coverage. Without them, the indicator shrinks to $-2.8\%$ and $-3.2\%$, neither significant, while the expansion estimate barely moves, to declines of 29% and 27%. Part of the rest comes from a change of survey, since the uninsured series switches from the Current Population Survey to the American Community Survey in 2013, the year before most states adopted. On an American Community Survey series for 2008–2019, the preferred estimates are declines of 24% unweighted and 22% weighted, still highly significant, and the indicator for the two years before

adoption is −2.7% and −2.8%, neither significant. About half of the difference from the full-sample estimates comes from the shorter window rather than from the change of survey. Allowing for state-specific linear trends in column G shrinks the full-sample estimates to declines of 22% and 19%, which remain highly significant. The negative binomial estimates are nearly identical to the unweighted OLS estimates.

In Figure 3, I estimate the divergence between expanding and non-expanding states over time. The figure plots coefficients from an event-study version of the model with state FE, year FE, region-by-year FE, and the poverty rate, unemployment rate, and log median household income as time-varying covariates. I allow for divergence from five years before adoption to five or more years after. Estimates are relative to six or more years before adoption.

Before adoption, coefficients for weighted and unweighted models are close to zero.[5] The decline begins in the year of adoption, and one year after adoption a state that expands Medicaid experiences an average estimated drop of 28% in its uninsured rate (27% weighted), relative to a state that has not expanded. The gap widens slightly thereafter, to between 34 and 35% by five years after adoption.

There may be concerns that Medicaid expansion increased Medicaid enrollment by drawing people out of other coverage and thus not reaching low-income populations. Panel A of Table 3 shows that Medicaid enrollment rises, on average, by 42% for expanding states relative to non-expanding states, and the robustness appendix reports every specification. Medicaid enrollment in adopting states was already rising somewhat faster before adoption. The pre-adoption event-study coefficients are not jointly significant, but the indicator for the two years before adoption is positive and significant (6.3% unweighted), and with state-specific linear trends the estimate is 31% (25% weighted), so I treat 42 to 44% as the upper end of the range. I also consider PHI enrollment with the same specifications. Expanding states, on average, experience a reduction of the PHI enrollment rate by 2.7%. One reason is mechanical. In non-expanding states, adults between 100 and 138% of the FPL qualified for subsidized private plans on the ACA Marketplaces, which these data count as private insurance, while in expanding states the same adults qualified for Medicaid. Some low-income individuals may also have had health insurance through their work but found Medicaid cheaper.

Combining these results, I find that the previously uninsured are the largest source of new Medicaid enrollees. Evaluated at the sample means in Table 1, a 42% increase in Medicaid enrollment is about 7.2 percentage points of the population, the 29% decline in the uninsured rate is about 3.5 percentage points, and the 2.7% decline in private enrollment is about 1.7 percentage points. The decline in uninsurance therefore accounts for about half of the rise in Medicaid enrollment, and the decline in private coverage for about a quarter. The remainder reflects, at least in part, the difference between administrative enrollment counts and survey-based measures of coverage, which I return to in the conclusion.

[5] Both coefficients dip slightly in the year before adoption (by about 5%). For the 2014 cohort that year is 2013, the first year of the American Community Survey series, and several states (CA, MN, DC) also enacted early coverage options before 2014.

I then test the identifying assumption directly by examining whether a type of health insurance enrollment that should not be affected by expansion, and that tracks general health insurance trends, is affected. If it is substantially affected when it shouldn't be, this would cast doubt on the model.

I thus examine how Medicaid expansion affects Medicare enrollment. While there may be some effect, it should be second-order at most. The Medicare estimate is small and insignificant both in the two-way FE benchmark of column A, at 0.6% unweighted and 1.0% weighted, and in my preferred model, at −0.8% unweighted and 0.6% weighted. Expansion therefore does not pick up general trends in health insurance, as measured by Medicare enrollment, which supports the identifying assumption. Because Medicare also covers disabled adults, whose enrollment can respond to local labor markets, I read this as a check on broad coverage trends rather than a sharp placebo. Table 3 collects the preferred estimates for all state-level outcomes.

### 5.2 Emergency Department Visits

Coverage is only half of the question. The other half is who pays for care, which I begin to answer with emergency department visits.

Expansion did not detectably change the total number of adult ED visits. These results are in Panel B of Table 3. The preferred estimates are 0.4% and −1.1%, with standard errors about twice as large or more, and as with Medicare enrollment, the two-way FE benchmark gives a similar null, at −0.5% unweighted and 1.7% weighted. The 95% confidence intervals rule out changes in total ED visits larger than about 6% in either direction, but not smaller ones. Even if the newly insured raised their ED use by 40%, as in Oregon (Taubman et al., 2014), total ED visits would rise by only a few percent, because they are a small share of all residents. The ED panel is also thinner than the coverage panel. It covers 23 states in 2006 and 36 in 2019, and with region-by-year FE the preferred estimates rest mainly on Kentucky, Maine, Iowa, Minnesota, Indiana, North Dakota, Montana, and Arkansas. I then examine whether payer type matters for changes in ED visits.

Panel B of Table 3 also shows my model's estimates for Medicaid-paid ED visits. States that expand Medicaid experience, on average, a 58% increase in Medicaid ED visits (53% weighted). This is expected, since more individuals have Medicaid. The same panel shows the corresponding estimates for self-pay/no-charge ED visits. On average, states that expand Medicaid experience a 44% decrease in self-pay/no-charge ED visits (42% weighted). One explanation is that patients who previously paid out of pocket, or did not pay at all, are now paying with Medicaid. The event studies in Figure 4 support this reading, as both series are flat before adoption and move sharply, in opposite directions, in the year of adoption. The explanation is further supported by the remaining payer types. Medicare ED visits do not move. Privately insured ED visits fall by 7.5% unweighted (significant at the 10% level) and 10.8% weighted, so the shift away from self-pay is not the only change in the payer mix, though it is by far the largest, and it points the same way as the modest decline in private enrollment. The two-way FE benchmark, which draws on nearly every reporting state, gives similar payer estimates, a 52% rise in Medicaid visits and a 37% fall in self-pay/no-charge visits unweighted. It is thus reasonable to conclude that states that expand Medicaid experience a

large shift from self-pay/no-charge payers to Medicaid payers in the emergency department, with no detectable change in the total. This is the shift Nikpay et al. (2017) find in the first year of expansion, and these estimates show it persisting through 2019.

Asthma offers a sharper version of the same test, because many asthma hospitalizations can be prevented with good primary care, so better access should show up as fewer of them. For asthma, the HCUP state data report adult care by payer only for inpatient stays. Medicaid-paid asthma stays rise modestly, by 19% unweighted (significant at the 10% level) and an insignificant 16% weighted, while uninsured asthma stays fall by 42 to 43% in both models. Total asthma stays do not change detectably, at −0.5% unweighted and 0.3% weighted, with standard errors of about 5%. As at the ED, expansion changed who paid for this care rather than how much of it was delivered. Uninsured asthma stays are a small count in many states, and the series covers only 368 state-years, so I read these estimates as consistent with the payer shift rather than as evidence about asthma care itself; the full tables are in the robustness appendix.

While there is a significant increase in Medicaid-paid ED visits, it is accompanied by a decrease in self-pay and no-charge visits of similar proportional size. This is a potential benefit for hospitals, which are required to treat emergency patients regardless of their ability to pay and which bear much of the cost when those patients cannot. Uncompensated costs strain hospital finances, especially for the rural and safety-net hospitals that treat many low-income patients. Increasing Medicaid enrollment has also raised Medicaid spending, and whether that makes expansion a net benefit to taxpayers depends on costs and benefits this paper does not measure. What the state data do establish is that the main change at the ED was in who paid, with no detectable change in how much care was delivered. I return to this at the hospital level below.

### 5.3 Rural Hospitals

The state-level results establish that expansion moved coverage and shifted who paid for emergency care. I now ask whether that shift reached the hospitals that had been absorbing the cost of treating uninsured patients. Rural hospitals are a natural place to look. They are often the only source of emergency care in their area, they treated a population that was disproportionately uninsured before 2014, and they operate on thin margins. The average rural hospital in my sample spends more on operations than it receives in patient revenue, with a mean operating margin of −8.4%, and breaks even only after other income such as local tax support. Table 4 describes the panel.

The hospital tables follow the structure of the state tables, with one row per outcome and one column per specification. Column 1 includes hospital and year FE, and column 2, my preferred model, replaces the year effects with region-by-year effects. Column 3 adds the county poverty rate, unemployment rate, and log median household income, and these controls remain in every later column. Column 4 weights each hospital by its 2011–2013 operating expense, so that it describes the average dollar of hospital spending rather than the average hospital, and column 5 adds state-specific linear trends. Column 6 adds to column 3 an indicator for the two years before adoption, whose coefficient is reported in the indented row beneath each outcome, and column 7 re-estimates column

6 without hospitals in the ten states that covered low-income adults before 2014.

Before turning to the estimates, I check whether rural hospitals in adopting and non-adopting states were on similar paths before expansion. Figure 5 plots the raw means for the 2014 cohort, and Figure 6 plots event studies with region-by-year FE and the county covariates. For eight of the nine outcomes, the pre-adoption coefficients are jointly insignificant, with $p$-values between 0.06 and 0.84. The exception is occupancy ($p = 0.047$), whose estimates are small and insignificant in every specification. The indicator for the two years before adoption in column 6 points the same way. It is small and insignificant for every outcome except log charity care and log discharges, where it is significant only at the 10% level.

I measure the payer-mix outcomes in logs because their dollar amounts are highly skewed. In 2013, about half of all charity care dollars at rural hospitals in expanding states sat at a tenth of those hospitals. As shares of operating expense, which is how uncompensated care is usually reported, the three outcomes pass the same pre-adoption test, and the robustness appendix reports them. Measuring treatment by the share of each cost report that falls after expansion matters here. Under a simple indicator for years at or after expansion, the reference year already contains post-expansion months for about two in five hospitals in adopting states, and charity care as a share of operating expense then appears to be falling before adoption ($p = 0.01$).

Operating margin passes the joint test ($p = 0.23$), but its earliest coefficient, which pools four or more years before adoption and so draws on 2011 for the 2015 cohort, 2011 and 2012 for the 2016 cohort, and longer stretches for Maine and Virginia, is 2.6 percentage points, about as large as the estimated effect. The coefficients two and three years before adoption are within 0.6 points of zero.

### 5.3.1 Payer Mix

Expansion reached the communities these hospitals serve, and it cut the uncompensated care they carried. Results for payer mix are in Table 5. The uninsured rate among low-income adults in a hospital's county falls by 8.1 percentage points against a 2011–2013 mean of 34.8% in expanding states, a decline of nearly a quarter, and the estimate ranges from 6.7 to 9.3 points across specifications.[6] The cost of charity care for uninsured patients falls by 59%, and the estimate ranges from 56 to 67% across specifications. The event study shows little movement before adoption, a decline of about 27% in the first cost report with post-expansion months, which often covers only part of a year, and a decline of 59 to 64% in the years after. Charity care falls by more than twice as much, in proportional terms, as the low-income uninsured rate. One explanation is that the people who gained coverage were those with the most hospital care to pay for. The decline does not appear to be an artifact of reporting. Critical access hospitals, 59% of the sample, are paid by Medicare on the basis of cost and receive no Medicare uncompensated care payments, so they had no reason to change how they report charity care when CMS began using these reports to allocate

[6] SAHIE estimates are model-based and combine the American Community Survey with administrative records that include county Medicaid and CHIP enrollment, so this row partly reflects enrollment directly. I read it as a check that coverage moved where these hospitals are.

those payments in fiscal year 2018. Their charity care falls by 53%, against 67% at the other rural hospitals. Non-Medicare bad debt, the part of uncompensated care that comes from patients who were billed and did not pay, falls by 29%, and by 21 to 31% in the other specifications.

Not all of this is a gain for hospitals. The unreimbursed cost of treating Medicaid patients rises by 20%, though this estimate is significant only at the 10% level and is insignificant in three of the seven specifications. Measured as a share of operating expense, it rises by 0.74 percentage points, significant at the 1% level, with no sign of a pre-trend. Newly covered patients now pay through Medicaid, and Medicaid pays less than the cost of their care, so part of what hospitals used to lose to uncompensated care they now lose to Medicaid shortfalls instead. Applied to the average rural hospital in an expanding state in 2013, counting a hospital that reports none of an item as zero, the three estimates imply about $410,000 less charity care, $400,000 less bad debt, and $260,000 more in Medicaid shortfalls a year, in 2019 dollars. Netting the three, expansion improved the average hospital's position by roughly $545,000 a year (95% confidence interval $320,000 to $770,000), or about 1.0% of operating expense.

### 5.3.2 Margins and Volume

Operating margins rose after expansion, but this is the least secure estimate in the paper. Operating conditions are in Table 6. Operating margin rises by 2.23 percentage points against a sample mean of −8.38%, significant at the 1% level, and the estimate ranges from 1.29 to 3.57 points across specifications. It is weakest once the indicator for the two years before adoption is added, at 1.61 and significant only at the 10% level, and loses significance at 1.29 when the early-coverage states are also dropped. I read this estimate as suggestive. Its earliest pre-adoption coefficient is as large as the estimate itself, and under the sensitivity analysis of Rambachan and Roth (2023) its confidence interval includes zero once post-adoption departures from parallel trends are allowed to be even a small fraction of the largest pre-adoption change. Two pieces of evidence still point toward a real improvement. For the 2014 cohort alone, whose pre-adoption coefficients pass the test, the margin rises by 2.36 points (standard error 0.80). Adding year effects interacted with baseline hospital characteristics, which lets government hospitals, more common in non-adopting states, follow their own path, leaves the estimate at 2.10. Taken at face value, the estimate closes about a quarter of the gap between the average rural hospital and break-even, and it is larger than the 1.1 points Blavin (2016) finds for all hospitals in the first year of expansion. At about $1.2 million a year for the average hospital, though, it is about twice the net payer-mix relief computed above, and the within-state comparison in section 6 does not support a payer-mix explanation for it, so the payer-mix estimates are the firmer measure of what expansion did for hospital finances.

Expansion did not detectably change how much inpatient care rural hospitals delivered. Log discharges fall by an insignificant 2.7% in the preferred specification and by about 5% in columns 6 and 7, significant only at the 10% level, where the indicator for the two years before adoption is also negative. Occupancy is indistinguishable from zero in every specification.

Revenue and cost point the same way. Net patient revenue rises by an insignificant 1.8%, and

operating expense is essentially unchanged, at −0.1%. Neither estimate is precise, but together they suggest that the margin gain came from revenue growing slightly relative to costs rather than from treating more patients. With state linear trends, the revenue estimate is 2.9% and significant at the 1% level, and the margin estimate also survives. Read alongside the state-level finding that Medicaid ED visits rose while self-pay and no-charge visits fell by a comparable amount, this is the pattern one would expect if expansion paid for care hospitals were already providing rather than inducing additional care.

These estimates should be read against the environment in which they arrived. Across the whole sample, without conditioning on expansion status, the median rural hospital discharged about 830 inpatients in 2011 and about 540 in 2019, and mean occupancy fell from 37 to 34%. Rural inpatient care was contracting for reasons expansion does not act on, so the absence of a detectable effect on volume says that expansion did not reverse that decline, not that volume was stable.

Taken together, the hospital results confirm the state-level picture from the other side of the transaction. Expansion lowered the uninsured rate in rural hospitals' counties, and substantially reduced the charity care and bad debt those hospitals carried, and it did so with little change in the volume of care they delivered. Part of the gain came back as Medicaid shortfalls. Operating margins may have improved as well, though that estimate is less secure.

## 6 Robustness

A separate robustness appendix reports the checks summarized here.

*Staggered adoption.* The stacked difference-in-differences estimates, which compare each adoption cohort only with states that have not yet adopted, show no divergence three and two years before adoption, relative to the year before, for the uninsured rate (3.4% and 3.7%, neither significant), Medicaid ED visits, or self-pay ED visits, and a divergence of only 1% for PHI enrollment two years before, significant at the 10% level. Post-adoption effects grow over time, from −13% in the year of adoption to −25% two years later for the uninsured rate. The Callaway and Sant'Anna (2021) estimates, which use never-adopting states as controls, are smaller than the two-way FE estimates but have the same sign and significance, at −19% for the uninsured rate, 25% for Medicaid enrollment, 51% for Medicaid ED visits, and −30% for self-pay ED visits. The gap does not come from region-by-year effects, since the two-way FE estimate without them is a 28% decline. For the uninsured rate it comes from the earlier years of the sample and the 2013 change of survey, which this estimator avoids by comparing each cohort with the year before adoption, and from its covariates.

*Survey source.* On an American Community Survey series for 2008–2019, the preferred estimates are declines of 24% unweighted and 22% weighted, as discussed above, and the post-adoption event-study coefficients are nearly the same as those from the spliced series over the same years.

*Comparison states.* Wisconsin did not expand, but it covered childless adults up to 100% of the FPL from April 2014 under a waiver, and it is the only Great Lakes state that had not expanded

by 2019. Dropping it raises the preferred estimate to a 31% decline in the uninsured rate (28% weighted). Dropping the three regions in which every state had expanded by 2019 leaves the estimate at a 28% decline (26% weighted) and the Medicaid ED estimate at a 49% rise. Adoption also does not predict the unemployment rate or median household income, and it predicts a poverty rate 0.2 to 0.4 percentage points higher, significant only when weighted, so the controls in column C change little.

*Randomization inference.* I reassign the actual set of adoption years to randomly chosen states 1,000 times, shifting the whole block of adoption years by a common number of years, and re-estimate the preferred model on each placebo assignment. For the uninsured rate, none of the 1,000 placebo estimates is as negative as the actual estimate, a 29% decline. The placebo distributions for the other outcomes are in the appendix.

*Hospital sample.* Rural status in the Provider of Services file is recorded annually, and about 7% of hospitals change status at least once. Expansion does not predict leaving the rural sample (a coefficient of 0.004 with a standard error of 0.010), and I fix each hospital's status at its 2011–2013 value in any case. Hospitals in non-adopting states were more likely to stop filing before 2019, at 8.7% against 3.7%, more than half of them through closure, but restricting the sample to hospitals that filed in every year leaves charity care at a 59% decline and the margin at 1.98 points. Expansion also did not change whether a hospital reported any charity care (a change of −1.3 percentage points, with a standard error of 1.4), so the log estimates are not driven by hospitals dropping out of them. Dropping the three regions in which every state had expanded by 2019 leaves every payer-mix and margin estimate within a point of the preferred one.

*Hospital pre-trends.* The appendix re-estimates every hospital outcome with county demographics, with year effects interacted with baseline hospital characteristics, with year effects interacted with the hospital's 2011 outcome, with state linear trends, and with state trends estimated on years before adoption and projected forward. None of these changes the sign or significance of the charity care or bad debt estimates in logs, which range from 56 to 61% and from 21 to 28%, though the Medicaid shortfall estimate becomes imprecise once pre-adoption state trends are removed. Under the sensitivity analysis of Rambachan and Roth (2023), the decline in log charity care remains significant until post-adoption departures from parallel trends reach 0.8 times the largest change between adjacent pre-adoption years, and at least twice that change once the pooled earliest period is left out. The operating margin estimate does not survive any such departure in the specification with county covariates, and survives only up to 0.3 times without the pooled earliest period.

*Within-state exposure.* The hospital design above treats every hospital in an adopting state as equally affected. The appendix also compares hospitals in the same state and year that differed in their uncompensated care burden before expansion, in the spirit of Finkelstein (2007), with payer mix measured as shares of operating expense (because the exposure measure is itself a share). A hospital one standard deviation higher in pre-period burden saw an additional decline in charity care for the uninsured of 1.74 percentage points, but also an additional rise in unreimbursed Medicaid cost of 0.98 points, and its operating margin rose by 2.30 points less than other hospitals' (significant

at the 5% level). The relief and the Medicaid shortfall that offset it were therefore both concentrated at the hospitals that had carried the most uncompensated care. The net payer-mix gradient is small next to the margin gradient and, if anything, favors the high-burden hospitals, so the payer-mix shift does not explain why their margins rose less, which is one more reason I treat the margin estimate as suggestive.

# 7 Conclusion

This paper follows Medicaid expansion from insurance coverage to the emergency department and then to the finances of rural hospitals. From 2014 to 2024, most states expanded Medicaid coverage to lower-income adults, and there are several reasons these individuals might not participate. They may not be aware of their eligibility; they may not enroll due to worries about overcrowded hospitals; they may refrain from enrolling for political or personal reasons.

Results presented in this paper, though, indicate that these potential deterrents have not stopped lower-income individuals from enrolling in Medicaid. I estimate a decline of roughly 26 to 29% in the uninsured rate (22 to 24% on the American Community Survey alone), which is largely driven by increased Medicaid enrollment among the lower-income population gaining coverage, with a smaller shift out of private insurance.

Total emergency department visits do not change detectably in states that expand Medicaid. This is also true of ED visits by Medicare patients. However, Medicaid-paid ED visits increase by more than half on average, offset by a decrease of more than 40% in self-pay/no-charge visits. While total ED visits show no detectable change, the shift toward Medicaid payers means hospitals absorb less uncompensated care, which is confirmed when shifting focus to the hospital level. At rural hospitals, charity care for the uninsured falls by more than half and bad debt by more than a quarter, with little change in volume. Part of that relief is offset by Medicaid shortfalls, and operating margins may improve, though that estimate is less secure. At the emergency department and at rural hospitals, then, much of what expansion paid for was care that was already being delivered and previously went unpaid, which is consistent with the estimate of Finkelstein et al. (2019) that about 60% of Medicaid spending in Oregon was a transfer to the providers of uncompensated care.

A back-of-the-envelope calculation puts these estimates in levels. I apply the population-weighted estimates, which describe the average resident, to 2013 levels in the 33 states and the District of Columbia that expanded by 2019. Those states had 26.5 million uninsured residents in 2013, so a 26% decline implies about 6.9 million fewer uninsured people. Their Medicaid enrollment was 40.4 million, so a 32% increase implies about 13.1 million additional enrollees, and their private enrollment of 127 million implies about 3.8 million fewer privately insured people. The enrollment counts and the survey-based uninsured rate measure coverage differently, so these figures need not balance, but they again point to the previously uninsured as the main source of new Medicaid enrollees.

Administrative data give the cost side directly. In fiscal year 2019, Medicaid spending on the

expansion group in these states totaled $93.8 billion, $84.9 billion of which was paid by the federal government (Guth et al., 2021). At the emergency department, applying the estimates to 2013 visit rates in the 17 expanding states that report to HCUP, scaled to the population of all expanding states, implies about 6.5 million additional Medicaid-paid visits a year, against about 4.3 million fewer self-pay/no-charge visits and 1.8 million fewer privately insured visits. Nearly all of the gain in Medicaid visits is therefore accounted for by visits that would otherwise have been paid out of pocket, left unpaid, or paid by private insurance.

The hospital estimates can be scaled the same way. In 2013 the average rural hospital in a state that expanded by 2019 had operating expenses of about $55 million and carried about $690,000 in charity care for uninsured patients, $1.4 million in non-Medicare bad debt, and $1.3 million in unreimbursed Medicaid cost, in 2019 dollars, counting a hospital that reports none of an item as zero. The estimated declines of 59% and 29% remove about $0.8 million a year of charity care and bad debt per hospital, and the 20% rise in Medicaid shortfalls gives back about $0.3 million, for net relief of roughly $0.55 million (95% confidence interval $0.32 million to $0.77 million), or 1.0% of operating expense. The less secure margin estimate implies more, since 2.23 percentage points of the average hospital's $53 million in net patient revenue is about $1.2 million. Summed across the roughly 1,200 rural hospitals in these states, the net payer-mix relief is about two-thirds of a billion dollars a year, or about 0.7% of fiscal year 2019 spending on the expansion group. Rural hospitals deliver a small share of hospital care, so this is far from the total amount of uncompensated care the new spending replaced. It does show that the relief was meaningful for hospitals operating on thin margins and small relative to the program. Because these calculations apply average effects to a single year's levels and set 2013 hospital levels against 2019 program spending, they convey orders of magnitude rather than precise totals.

Two questions remain. The first is whether the financial relief changes which rural hospitals survive. Closures remain too rare, and too recent, to answer that with the design used here, but the margins studied in this paper are the place to watch as the post-expansion period lengthens. The second is how expansion affects state budgets. Expanding states now pay 10% of the cost of the expansion population, a share that rose from zero in 2016 to 10% in 2020. In fiscal year 2019, states paid $8.9 billion of the $93.8 billion spent on the expansion group (Guth et al., 2021). The relief estimated here goes to hospitals rather than to state treasuries, so whether expansion pays for itself from a state's perspective depends on offsets this paper does not measure, such as lower state and local spending on uncompensated care.

These questions are now pressing. The 2025 federal budget reconciliation law left the 90% federal match in place, but it requires states to condition coverage for expansion adults on work requirements by the end of 2026, to recheck their eligibility every six months, and, starting in fiscal year 2028, to rely less on the provider taxes many states use to fund their share (KFF, 2025). The Congressional Budget Office projects that the law will raise the number of uninsured by about 10 million. If the effects estimated here run in reverse as coverage falls, as Garthwaite et al. (2018) find they did when Tennessee disenrolled adults from Medicaid in 2005, charity care and bad debt

would return to the rural hospitals that expansion relieved of them. The same law funds a $50 billion Rural Health Transformation Program for fiscal years 2026 through 2030, and the estimates here offer one benchmark for the relief rural hospitals stand to lose.

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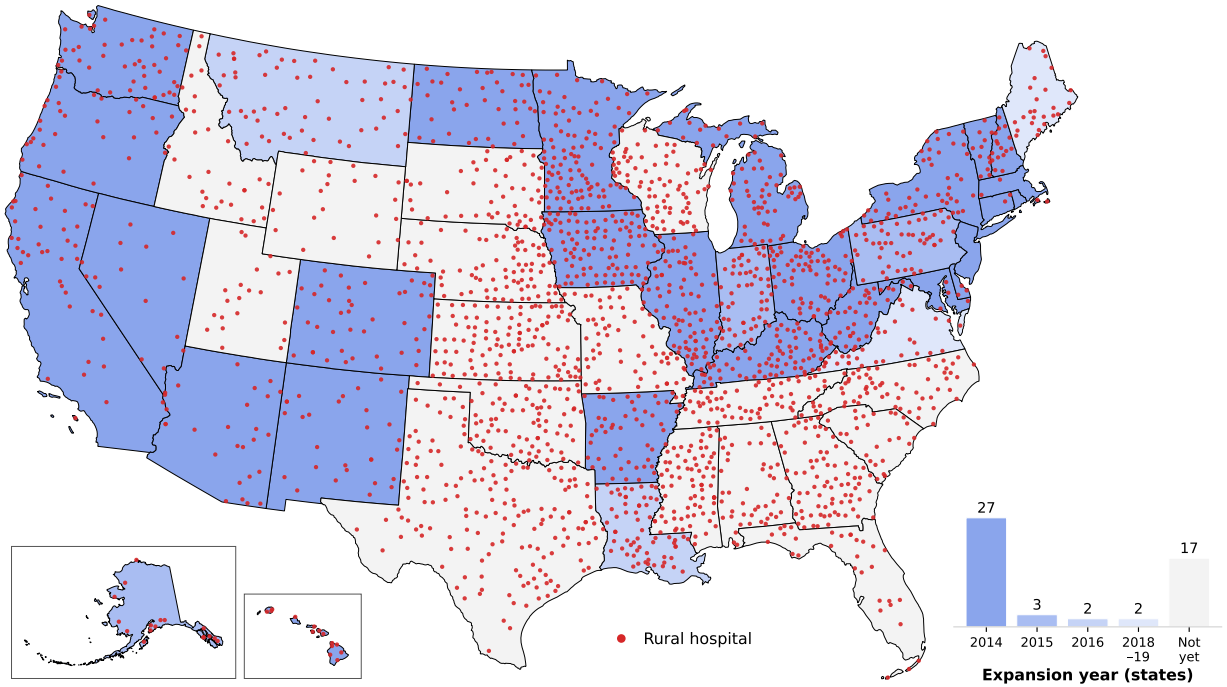
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# Figures and Tables

**Figure 1: Medicaid Expansion Effective Year, by State, and Rural Hospitals**



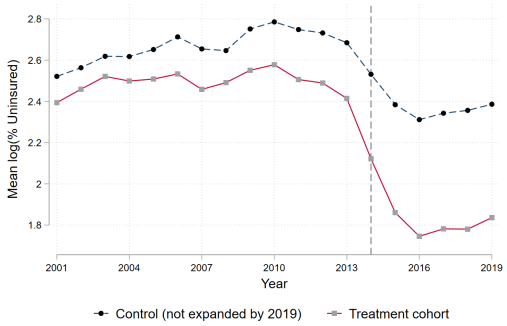

NOTES: States are shaded by the year Medicaid expansion took effect, through 2019; states that had not expanded by 2019 are gray. The histogram counts states, including the District of Columbia, by expansion year. Each point is a rural hospital in the analysis sample, placed at its ZIP code centroid. Expansion dates are from the Kaiser Family Foundation.

**Table 1: Descriptive Statistics, State Panel**

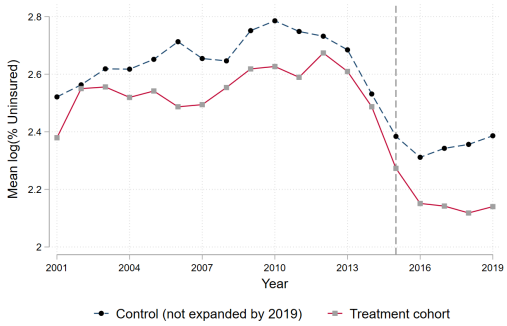
| | Unweighted | | Weighted by population | | |
|---|---|---|---|---|---|
| | Mean | S.D. | Mean | S.D. | $N$ |
| **Panel A. Insurance coverage and enrollment (% of population)** | | | | | |
| Uninsured rate | 12.02 | 4.37 | 13.17 | 4.97 | 969 |
| Medicaid enrollment | 17.19 | 6.34 | 17.97 | 5.75 | 969 |
| Medicare enrollment | 16.00 | 2.74 | 15.50 | 2.60 | 969 |
| Private insurance enrollment | 65.39 | 6.61 | 63.83 | 6.52 | 969 |
| **Panel B. Adult emergency department visits (per 100 residents)** | | | | | |
| Total | 26.81 | 5.06 | 26.56 | 4.81 | 403 |
| Medicaid | 6.23 | 2.71 | 6.36 | 2.58 | 403 |
| Medicare | 6.58 | 1.42 | 6.26 | 1.38 | 403 |
| Private insurance | 9.11 | 1.68 | 8.62 | 1.77 | 403 |
| Self-pay / no charge | 4.89 | 2.80 | 5.32 | 2.77 | 403 |
| **Panel C. Asthma inpatient stays (per 100,000 residents)** | | | | | |
| Medicaid | 14.82 | 7.49 | 15.97 | 9.03 | 472 |
| Uninsured | 9.02 | 3.93 | 8.66 | 4.05 | 368 |
| **Panel D. Covariates** | | | | | |
| Poverty rate (%) | 13.38 | 3.24 | 13.81 | 2.81 | 969 |
| Unemployment rate (%) | 5.60 | 2.01 | 6.00 | 2.09 | 969 |
| Median household income (2019 dollars) | 60,556 | 9,849 | 61,555 | 9,022 | 969 |

NOTES: Means and standard deviations across state-year observations for the 50 states and the District of Columbia, unweighted and weighted by the state's average population over the sample period. Coverage, enrollment, and the covariates span 2001–2019. The uninsured rate comes from the Current Population Survey through 2012 and the American Community Survey from 2013. ED visits come from HCUP for 36 states in an unbalanced panel for 2006–2019 and exclude Illinois, and asthma inpatient stays come from HCUP and span 2003–2017. Enrollment is CMS National Health Expenditure Accounts enrollment of all ages over Census population.

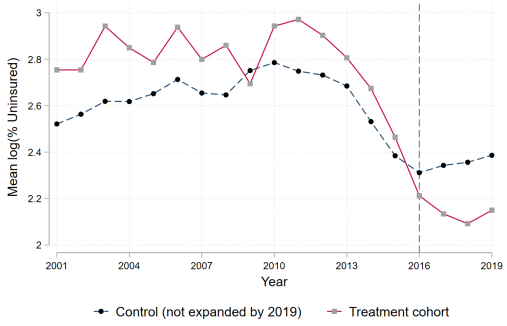
**Figure 2: Log Uninsured Rate Before and After Adoption of Medicaid Expansion, by Year of Adoption**



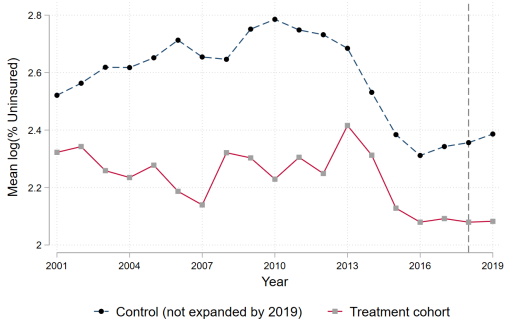

**(a) States adopting in 2014**



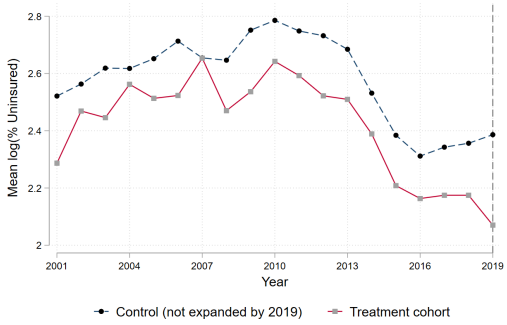

**(b) States adopting in 2015 (AK, IN, PA)**




**(c) States adopting in 2016 (LA, MT)**




**(d) States adopting in 2018 (ME)**




**(e) States adopting in 2019 (VA)**

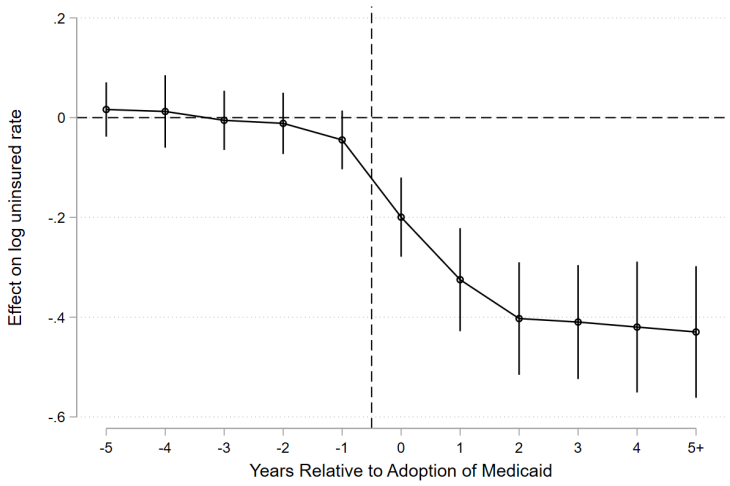
NOTES: Each panel plots the unweighted cross-state mean of the log uninsured rate by calendar year, separately for the indicated adoption cohort (treatment) and for states that had not adopted Medicaid expansion by 2019 (control). The vertical dashed line marks the cohort's adoption year. The 2014 cohort is AZ, AR, CA, CO, CT, DE, DC, HI, IL, IA, KY, MD, MA, MI, MN, NV, NH, NJ, NM, NY, ND, OH, OR, RI, VT, WA, and WV.

**Table 2: Log Uninsured Rate — Difference-in-Differences Estimates**

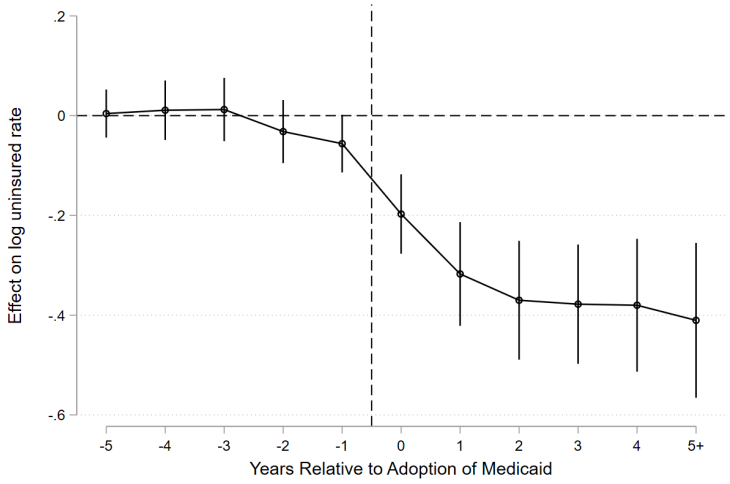
| | (A) | (B) | (C) | (D) | (E) | (F) | (G) |
|---|---|---|---|---|---|---|---|
| **Panel A: Unweighted OLS on log outcome** | | | | | | | |
| Med. Exp. | -0.3319*** | -0.3418*** | -0.3462*** | -0.3581*** | -0.3483*** | -0.3425*** | -0.2548*** |
| | ( 0.0307) | ( 0.0385) | ( 0.0427) | ( 0.0455) | ( 0.0457) | ( 0.0439) | ( 0.0334) |
| 1–2y pre | | | | -0.0417* | -0.0279 | | |
| | | | | ( 0.0236) | ( 0.0245) | | |
| **Panel B: Weighted OLS on log outcome (population)** | | | | | | | |
| Med. Exp. | -0.3325*** | -0.3004*** | -0.3128*** | -0.3309*** | -0.3196*** | -0.3152*** | -0.2143*** |
| | ( 0.0378) | ( 0.0446) | ( 0.0449) | ( 0.0476) | ( 0.0537) | ( 0.0456) | ( 0.0358) |
| 1–2y pre | | | | -0.0634** | -0.0328 | | |
| | | | | ( 0.0299) | ( 0.0201) | | |
| **Panel C: Negative binomial on count outcome (population exposure)** | | | | | | | |
| Med. Exp. | -0.3288*** | -0.3408*** | -0.3447*** | -0.3561*** | -0.3465*** | -0.3406*** | -0.2569*** |
| | ( 0.0300) | ( 0.0349) | ( 0.0387) | ( 0.0410) | ( 0.0399) | ( 0.0398) | ( 0.0288) |
| 1–2y pre | | | | -0.0401* | -0.0284 | | |
| | | | | ( 0.0209) | ( 0.0213) | | |
| State FE | Y | Y | Y | Y | Y | Y | Y |
| Year FE | Y | Y | Y | Y | Y | Y | Y |
| Region × year | | Y | Y | Y | Y | Y | Y |
| Poverty, unemployment, log median income | | | Y | Y | Y | Y | Y |
| 1–2y pre indicator | | | | Y | Y | | |
| Excludes early-coverage states | | | | | Y | | |
| Log Medicare enr. | | | | | | Y | |
| State linear trends | | | | | | | Y |
| Observations | 969 | 969 | 969 | 969 | 779 | 969 | 969 |

NOTES: Each column reports the coefficient on the Medicaid-expansion treatment variable (the share of the year after expansion) and, where included, the 1–2 year pre-treatment indicator from a separate state-year panel regression of Log Uninsured Rate on the listed covariates. Column (A) includes state fixed effects and year fixed effects. (B) adds region × year interactions. (C) adds the state poverty rate, unemployment rate, and log median household income, which remain in (D) through (G). (D) further adds an indicator equal to one in the two years immediately preceding adoption. (E) re-estimates (D) without the ten states that covered low-income adults before 2014 (CA, CT, DC, MN, NJ, and WA through early ACA expansions; DE, MA, NY, and VT through prior coverage). (F) replaces the pre-adoption indicator with the log Medicare enrollment rate. (G) adds state-specific linear time trends in lieu of the controls in (D) through (F). All specifications cluster standard errors at the state level (in parentheses). Panel A reports unweighted OLS on the log outcome; Panel B weights by the state's average population over the sample period. Panel C reports a negative-binomial regression of the underlying count on the same covariates, using state population as the exposure variable; coefficients are log incidence-rate ratios and so are directly comparable to the OLS-on-log estimates. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

**Figure 3: Event-Study Coefficients for the Log Uninsured Rate**




**(a) Unweighted OLS**




**(b) Weighted OLS**

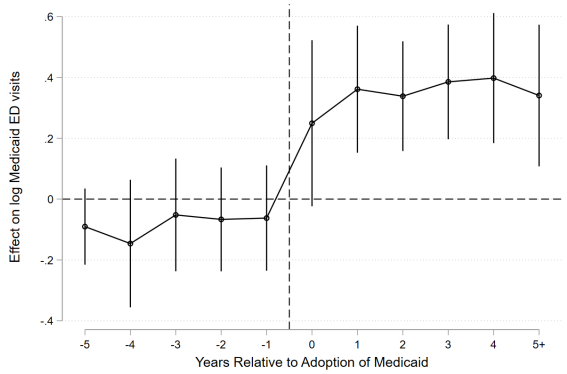
NOTES: Each panel plots the coefficients on indicators for years relative to a state's adoption of Medicaid expansion from a single regression that also includes state FE, year FE, BEA region-by-year FE, and the state poverty rate, unemployment rate, and log median household income. The reference category is six or more years before adoption, together with states that had not adopted by 2019. The vertical dashed line marks adoption; vertical bars are 95% confidence intervals based on standard errors clustered by state.

**Table 3: Effect of Medicaid Expansion on State Outcomes, Preferred Specification**

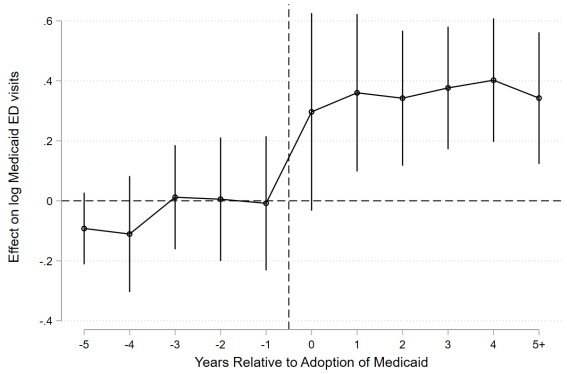
| | Unweighted OLS | | | Weighted by population | | |
|---|---|---|---|---|---|---|
| Log outcome | Coef. | (S.E.) | % change | Coef. | (S.E.) | % change |
| **Panel A. Insurance coverage (2001–2019, 969 state-years)** | | | | | | |
| Uninsured rate | $-0.342^{***}$ | (0.038) | −29.0 | $-0.300^{***}$ | (0.045) | −25.9 |
| Medicaid enrollment rate | $0.348^{***}$ | (0.041) | 41.7 | $0.280^{***}$ | (0.037) | 32.3 |
| Private insurance enrollment | $-0.027^{***}$ | (0.009) | −2.7 | $-0.030^{***}$ | (0.008) | −3.0 |
| Medicare enrollment (placebo) | −0.009 | (0.014) | −0.8 | 0.006 | (0.010) | 0.6 |
| **Panel B. Adult ED visits by expected payer (36 states, 403 state-years)** | | | | | | |
| Total | 0.004 | (0.025) | 0.4 | −0.011 | (0.022) | −1.1 |
| Medicaid | $0.456^{***}$ | (0.082) | 57.7 | $0.426^{***}$ | (0.094) | 53.1 |
| Self-pay / no charge | $-0.574^{***}$ | (0.199) | −43.7 | $-0.552^{**}$ | (0.265) | −42.4 |
| Medicare | −0.001 | (0.030) | −0.1 | 0.016 | (0.030) | 1.6 |
| Private insurance | $-0.078^{*}$ | (0.043) | −7.5 | $-0.115^{**}$ | (0.043) | −10.8 |

NOTES: Each row reports the coefficient on the Medicaid-expansion treatment variable from a separate regression of the indicated log outcome on state fixed effects, year fixed effects, and BEA-region-by-year fixed effects (column B of the full tables), and the implied percent change, $100 \times (e^{\beta} - 1)$. The treatment variable equals the share of the year in which the state's expansion was in effect. The ED sample is an unbalanced panel of the 36 states that report adult visits by payer to HCUP for some or all of 2006–2019, and it excludes Illinois because of gaps in its HCUP reporting. Standard errors clustered by state. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

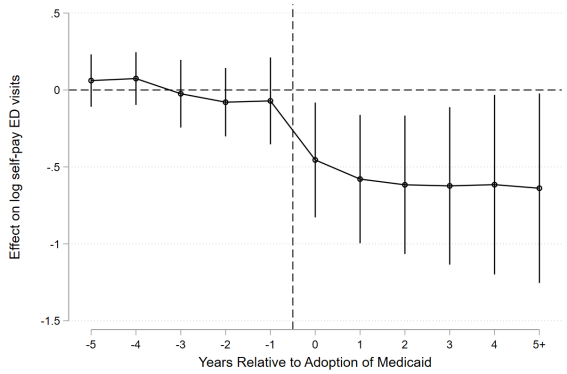
## Figure 4: Event-Study Coefficients for ED Visits by Payer



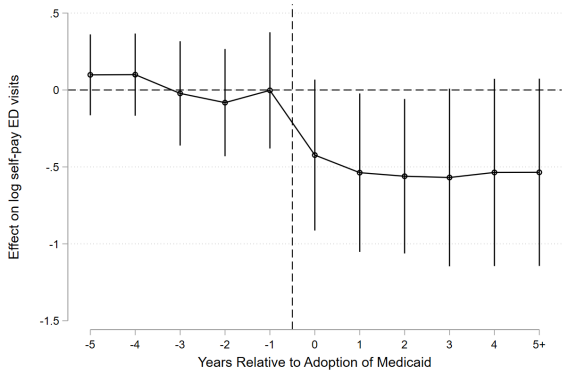

**(a) Medicaid ED visits, unweighted**




**(b) Medicaid ED visits, weighted**




**(c) Self-pay/no-charge ED visits, unweighted**




**(d) Self-pay/no-charge ED visits, weighted**

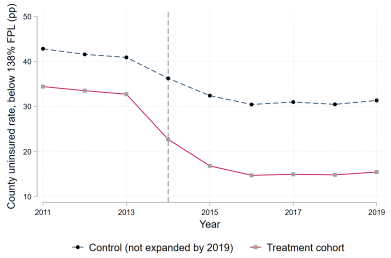
NOTES: As in Figure 3, except the outcomes are the logs of adult Medicaid-paid and self-pay/no-charge ED visits per resident, and Illinois is excluded.

**Table 4: Descriptive Statistics, Rural Hospital Panel**

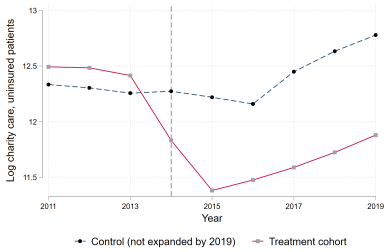
| | 2011–2019 | | 2011–2013 mean | | |
|---|---|---|---|---|---|
| | Mean | S.D. | Expansion | Non-expansion | *N* |
| **Panel A. Coverage and payer mix (% of operating expense unless noted)** | | | | | |
| County uninsured rate, adults below 138% FPL | 29.18 | 11.93 | 34.80 | 41.75 | 20,058 |
| Charity care for uninsured patients | 1.52 | 2.42 | 1.61 | 1.72 | 18,143 |
| Non-Medicare bad debt | 3.69 | 3.41 | 3.46 | 5.01 | 19,248 |
| Unreimbursed Medicaid cost | 3.75 | 3.92 | 3.52 | 3.01 | 13,596 |
| **Panel B. Margins, volume, and cost** | | | | | |
| Operating margin | -8.39 | 20.30 | -6.00 | -9.80 | 19,721 |
| Inpatient discharges | 1,550 | 2,395 | 1,701 | 1,606 | 20,011 |
| Occupancy rate | 34.79 | 18.40 | 37.71 | 34.00 | 20,002 |
| Net patient revenue (millions of dollars) | 51.04 | 73.04 | 52.91 | 41.04 | 19,739 |
| Operating expense (millions of dollars) | 52.12 | 71.69 | 53.99 | 41.18 | 20,033 |
| **Panel C. Hospital characteristics, fixed at 2011–2013 (% unless noted)** | | | | | |
| Beds | 58.94 | 68.01 | 58.99 | 58.38 | 20,138 |
| Critical access hospital | 58.91 | 49.20 | 62.45 | 53.75 | 20,139 |
| Nonprofit | 53.92 | 49.85 | 62.08 | 43.92 | 20,066 |
| For-profit | 11.73 | 32.18 | 9.69 | 15.00 | 20,066 |
| Government | 34.35 | 47.49 | 28.23 | 41.09 | 20,066 |
| Only acute hospital in its county | 61.91 | 48.56 | 54.87 | 67.83 | 20,139 |
| **Panel D. County covariates (% unless noted)** | | | | | |
| County poverty rate | 16.66 | 6.15 | 16.88 | 18.69 | 20,053 |
| County median household income (dollars) | 49,395 | 10,124 | 48,894 | 45,852 | 20,053 |
| County unemployment rate | 5.98 | 2.71 | 8.14 | 7.65 | 20,053 |

NOTES: Hospital-year observations for rural critical access and short-term acute hospitals that filed a cost report in 2011–2013. Expansion means the state expanded Medicaid by 2019. Margins and occupancy are in percent, and dollar amounts are in 2019 dollars.

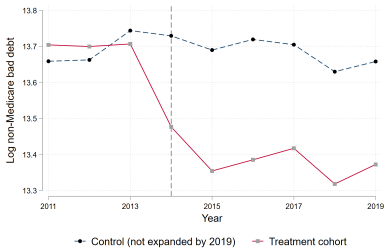
## Figure 5: Rural Hospital Outcomes Before and After Adoption, 2014 Cohort



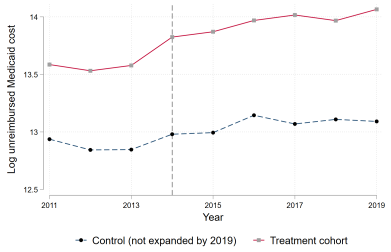

**(a) County uninsured, below 138% FPL**



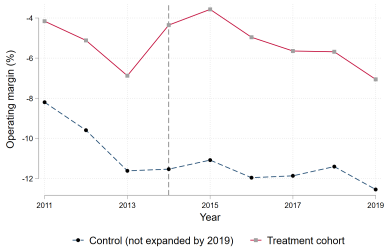

**(b) Log charity care, uninsured**



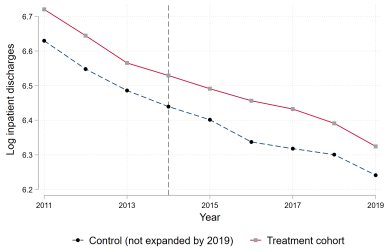

**(c) Log non-Medicare bad debt**



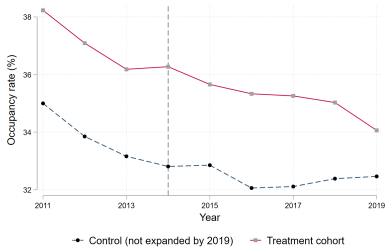

**(d) Log unreimbursed Medicaid cost**



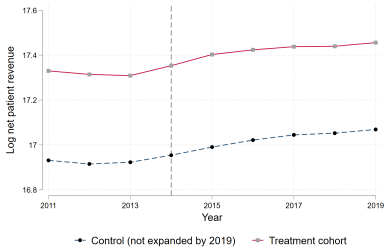

**(e) Operating margin**



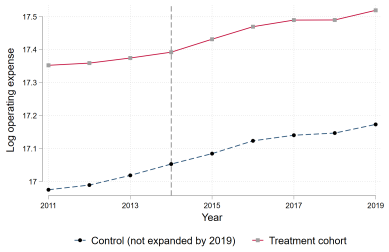

**(f) Log inpatient discharges**




**(g) Occupancy rate**




**(h) Log net patient revenue**




**(i) Log operating expense**

NOTES: Each panel plots the mean of the outcome across rural hospitals in states that expanded in 2014 (treatment) and in states that had not expanded by 2019 (control). The vertical dashed line marks 2014. The figures for the 2015, 2016, 2018, and 2019 cohorts are in the robustness appendix.

## Table 5: Medicaid Expansion and Rural Hospitals: Payer Mix

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | Mean |
|---|---|---|---|---|---|---|---|---|
| County uninsured rate, adults <138% FPL | -7.677*** | -8.070*** | -7.979*** | -7.976*** | -6.688*** | -8.268*** | -9.254*** | 29.17 |
| | (1.037) | (1.143) | (1.125) | (1.037) | (1.044) | (1.151) | (1.026) | |
| 1–2y pre | | | | | | -0.408 | -0.336 | |
| | | | | | | (0.453) | (0.495) | |
| Log charity care, uninsured patients | -0.984*** | -0.891*** | -0.891*** | -0.884*** | -0.818*** | -0.966*** | -1.094*** | 12.18 |
| | (0.142) | (0.153) | (0.155) | (0.175) | (0.114) | (0.174) | (0.174) | |
| 1–2y pre | | | | | | -0.093* | -0.087 | |
| | | | | | | (0.053) | (0.061) | |
| Log non-Medicare bad debt | -0.307*** | -0.337*** | -0.325*** | -0.334*** | -0.238*** | -0.338*** | -0.365*** | 13.61 |
| | (0.051) | (0.046) | (0.045) | (0.048) | (0.051) | (0.051) | (0.058) | |
| 1–2y pre | | | | | | -0.015 | -0.014 | |
| | | | | | | (0.028) | (0.033) | |
| Log unreimbursed Medicaid cost | 0.206*** | 0.180* | 0.175* | 0.128 | 0.142** | 0.167 | 0.238 | 13.48 |
| | (0.076) | (0.099) | (0.098) | (0.103) | (0.054) | (0.120) | (0.150) | |
| 1–2y pre | | | | | | -0.010 | 0.016 | |
| | | | | | | (0.055) | (0.062) | |
| Hospital FE | Y | Y | Y | Y | Y | Y | Y | |
| Year FE | Y | | | | | | | |
| Region × year FE | | Y | Y | Y | Y | Y | Y | |
| County covariates | | | Y | Y | Y | Y | Y | |
| Weighted by 2011–2013 operating expense | | | | Y | | | | |
| State linear trends | | | | | Y | | | |
| 1–2y pre indicator | | | | | | Y | Y | |
| Excludes early-coverage states | | | | | | | Y | |

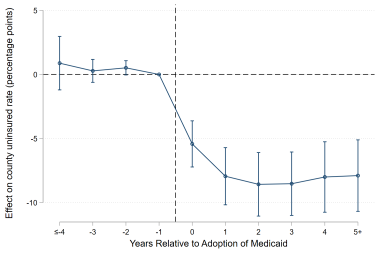
NOTES: The sample is rural critical access and short-term acute hospitals that filed in 2011–2013, observed 2011–2019. The county uninsured rate is the uninsured rate among adults below 138% of the federal poverty line in the hospital's county, in percent; the other three rows are in logs of 2019 dollars. Each cell is the coefficient on Post, the share of the hospital's cost-reporting period after the state's expansion date (for the county uninsured rate, the share of the calendar year), from a separate regression of the row outcome; hospitals in states that had not expanded by 2019 are controls. Column 2 is the preferred specification. Column 6 adds to column 3 an indicator for the two cost reports before the first report with post-expansion months (for the county uninsured rate, the two calendar years before expansion), whose coefficient is reported in the indented row, and column 7 re-estimates column 6 without hospitals in the ten states that covered low-income adults before 2014 (CA, CT, DC, MN, NJ, and WA through early ACA expansions; DE, MA, NY, and VT through prior coverage). The county covariates are the poverty rate, unemployment rate, and log median household income. The mean is for the column 3 sample, which has 13,443 to 20,044 hospital-years and 2,013 to 2,286 hospitals across 48 states, depending on the outcome. Standard errors clustered by state in parentheses. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

**Table 6: Medicaid Expansion and Rural Hospitals: Margins, Volume, and Cost**

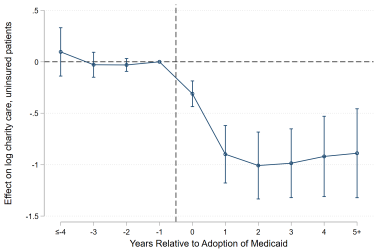
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | Mean |
|---|---|---|---|---|---|---|---|---|
| Operating margin | 2.452*** | 2.228*** | 2.172*** | 2.151** | 3.565*** | 1.613* | 1.287 | -8.38 |
| | (0.718) | (0.703) | (0.718) | (0.851) | (0.717) | (0.907) | (0.951) | |
| 1–2y pre | | | | | | -0.695 | -0.951 | |
| | | | | | | (0.596) | (0.590) | |
| Log inpatient discharges | 0.016 | -0.027 | -0.029 | -0.004 | 0.026 | -0.051* | -0.053* | 6.48 |
| | (0.026) | (0.019) | (0.019) | (0.017) | (0.017) | (0.027) | (0.031) | |
| 1–2y pre | | | | | | -0.028* | -0.031* | |
| | | | | | | (0.015) | (0.017) | |
| Occupancy rate | 0.013 | -0.367 | -0.389 | -0.781 | 0.611 | -0.717 | -0.877 | 34.80 |
| | (0.654) | (0.533) | (0.535) | (0.707) | (0.506) | (0.733) | (0.790) | |
| 1–2y pre | | | | | | -0.407 | -0.415 | |
| | | | | | | (0.418) | (0.489) | |
| Log net patient revenue | 0.026 | 0.018 | 0.014 | 0.026 | 0.029*** | 0.007 | 0.003 | 17.19 |
| | (0.021) | (0.015) | (0.014) | (0.020) | (0.010) | (0.020) | (0.022) | |
| 1–2y pre | | | | | | -0.009 | -0.011 | |
| | | | | | | (0.010) | (0.011) | |
| Log operating expense | 0.004 | -0.001 | -0.003 | 0.003 | -0.004 | 0.002 | 0.005 | 17.26 |
| | (0.022) | (0.012) | (0.011) | (0.012) | (0.008) | (0.017) | (0.019) | |
| 1–2y pre | | | | | | 0.006 | 0.009 | |
| | | | | | | (0.011) | (0.012) | |
| Hospital FE | Y | Y | Y | Y | Y | Y | Y | |
| Year FE | Y | | | | | | | |
| Region × year FE | | Y | Y | Y | Y | Y | Y | |
| County covariates | | | Y | Y | Y | Y | Y | |
| Weighted by 2011–2013 operating expense | | | | Y | | | | |
| State linear trends | | | | | Y | | | |
| 1–2y pre indicator | | | | | | Y | Y | |
| Excludes early-coverage states | | | | | | | Y | |

NOTES: The sample is rural critical access and short-term acute hospitals that filed in 2011–2013, observed 2011–2019. Operating margin and occupancy are in percent; net patient revenue and operating expense are in logs of 2019 dollars. Each cell is the coefficient on Post, the share of the hospital's cost-reporting period after the state's expansion date, from a separate regression of the row outcome; hospitals in states that had not expanded by 2019 are controls. Column 2 is the preferred specification. Column 6 adds to column 3 an indicator for the two cost reports before the first report with post-expansion months, whose coefficient is reported in the indented row, and column 7 re-estimates column 6 without hospitals in the ten states that covered low-income adults before 2014 (CA, CT, DC, MN, NJ, and WA through early ACA expansions; DE, MA, NY, and VT through prior coverage). The county covariates are the poverty rate, unemployment rate, and log median household income. The mean is for the column 3 sample, which has 19,638 to 19,936 hospital-years and 2,248 to 2,282 hospitals across 48 states, depending on the outcome. Standard errors clustered by state in parentheses. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

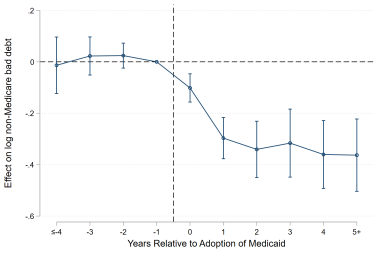
## Figure 6: Event-Study Coefficients for Rural Hospital Outcomes



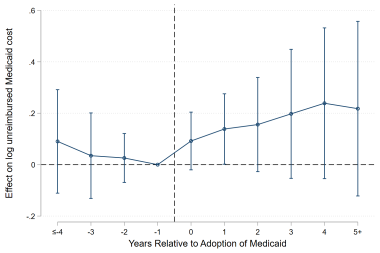

**(a) County uninsured, below 138% FPL**



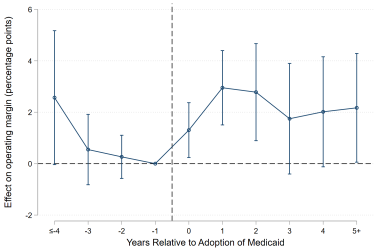

**(b) Log charity care, uninsured**



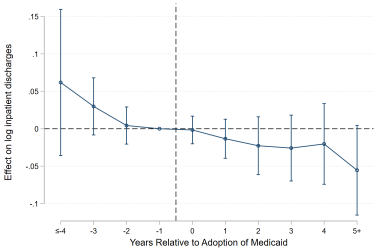

**(c) Log non-Medicare bad debt**



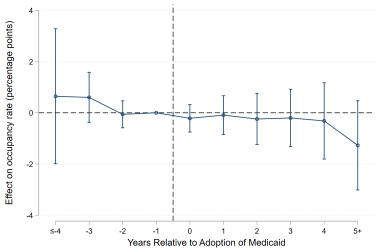

**(d) Log unreimbursed Medicaid cost**



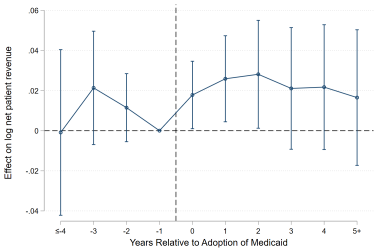

**(e) Operating margin**



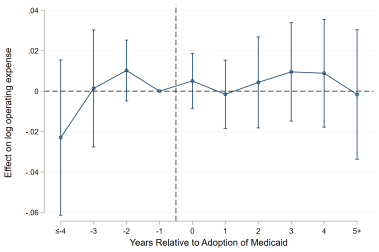

**(f) Log inpatient discharges**




**(g) Occupancy rate**




**(h) Log net patient revenue**




**(i) Log operating expense**

NOTES: Each panel plots the coefficients on indicators for years relative to the hospital's first cost report that includes months after its state's expansion (for the county uninsured rate, relative to the state's expansion year), omitting the year before, from a regression of the outcome on those indicators, hospital FE, BEA region-by-year FE, and the county poverty rate, unemployment rate, and log median household income. The earliest coefficient pools four or more years before adoption. Hospitals in states that had not expanded by 2019 are controls. Vertical bars are 95% confidence intervals based on standard errors clustered by state. The $p$-values for the joint test that the three pre-adoption coefficients are zero are 0.22, 0.27, 0.70, 0.84, 0.23, 0.15, 0.047, 0.20, and 0.06, in panel order.